

Homework (Jalapeno)

- Q1.** What is the GCF of 12 and 18?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12
- Q2.** The LCM of 9 and 15 is
(A) 15
(B) 45
(C) 90
(D) 135
(E) 225
- Q3.** In ancient Egypt, the Pyramids of Giza were built with stones weighing 12, 18, or 24 tons. What is the GCF of these weights?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12
- Q4.** The prime factorization of 42 is $2 \cdot 3 \cdot 7$. What is the LCM of 42 and 63?
(A) 42
(B) 63
(C) 126
(D) 252
(E) 315
- Q5.** A Mayan calendar has 15 days in a week and 20 weeks in a month. What is the LCM of 15 and 20?
(A) 15
(B) 20
(C) 60
(D) 75
(E) 100
- Q6.** The GCF of 24 and 30 is
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12
- Q7.** A Greek architect is designing a temple with columns spaced 8 meters apart and a hallway 12 meters wide. What is the GCF of 8 and 12?
(A) 1
(B) 2
(C) 4
(D) 8
(E) 12
- Q8.** The prime factorization of 48 is $2^4 \cdot 3$. What is the LCM of 48 and 75?
(A) 48
(B) 75
(C) 300
(D) 600
(E) 1200

Open Ended Questions (FRQ)

- Q9.** In the construction of the Great Wall of China, stones were placed 10 meters apart. If a watchtower was built every 25 meters, what is the LCM of 10 and 25? Show your work and explain your reasoning.
- Q10.** An ancient Babylonian farmer had 18 baskets of dates and 24 baskets of figs. If he wanted to pack the baskets in rows with the same number of baskets in each row, what is the GCF of 18 and 24? Show your work and explain your reasoning.

Chef's Tips

- Provide clear instructions for each problem
- Allow students to use calculators for checking
- Encourage students to show their work and explain their reasoning